

X is slower car of speed  $x$

Y is faster car of speed  $y$

In order for Y to pass X, the <sup>relative</sup> distance covered:  $(y-x)\frac{N}{y}$  needs to be less than the distance between Y and X. Thus, Y must have a start in the  $(y-x)\frac{N}{y}$  game before  $x$ . This occurs with a density? <sup>proportional to</sup>  $\frac{1}{xy}$ . Since X travels for  $\frac{N}{x}$  mins, we multiply by  $\frac{N}{x}$ :  $(\frac{1}{xy})(y-x)\frac{N^2}{x}$

Now, integrate over expected loss function:

$$E(a) = \int_a^a \int_x^a \frac{(y-x)zN^2}{xy} x^2 dy dx + \int_a^a \int_x^a \frac{(y-x)zN^2}{xy} (x-a)^2 dy dx$$
$$= \frac{zN^2}{18} \left[ a^3 - 9a + 3 + 6\ln(a) + a^3(17 + \ln(4)) + 18a^2(\ln(4) - 1) - 6a^2(a+6)\ln(a) - 36a + 3 \right]$$

$$\text{So solving } \frac{dE}{da} = 0 \Rightarrow a = 1.1771414168 \text{ mi/min.}$$